实 验 报 告

课程名称：OpenGauss DBMS **实验编号：二**

班级：22 计一 **姓名：刘星宇** **学号：320220941790**

指导老师：火克华 **实验日期：**2024/4/5 **学生电话：18717024225**

一. 实验内容

1. OpenGauss SQL 语言基础

1.1. DDL:数据定义语言

a) create:创建数据库对象；
b) drop:删除数据库对象；
c) alter:修改数据库对象；

1.2. 使用元命令修改数据库 DB_em 的表对象的属性

a) 列级：主键约束、非空约束、check 约束、default 约束、unique 约束
b) 表级：表间外键约束
c) 修改列的数据类型
d) 增加/删除列

1.3. DML 数据操作语言

a) Insert:向对象中插入数据；
b) Delete:删除对象中的数据；
c) Update:修改对象中的数据。

2. 创建数据库 DB_Bank，创建 5 个表对象；

初步体会数据库结构设计的要点，表结构设计如何满足第一范式、第二范式、第三范式。

二. 实验过程

1. 以 root 登录华为云 ECS,用 cloudshell 远程登录服务器。

```
Welcome to 4.19.90-2110.8.0.0119.oe1.aarch64

System information as of time:  Wed Apr  2 14:55:59 CST 2025

System load:    1.70
Processes:      147
Memory used:    10.8%
Swap used:      0.0%
Usage On:       15%
IP address:     192.168.0.53
Users online:   1
```

2. 登录 OS 管理员用户

```
[root@ecs-419c ~]# su


Welcome to 4.19.90-2110.8.0.0119.oe1.aarch64

System information as of time:  Wed Apr  2 15:28:59 CST 2025

System load:    0.16
Processes:      143
Memory used:    23.9%
Swap used:      0.0%
Usage On:       15%
IP address:     192.168.0.53
Users online:   1
```

3. 切换到 omm 用户，以操作系统互信用户 omm 登录数据库主节点

```
[root@ecs-3ee7~]# su - omm
```

```
[root@ecs-419c ~]# su - omm
Last login: Wed Apr  2 14:57:21 CST 2025 on pts/0
Last failed login: Wed Apr  2 15:01:33 CST 2025 on pts/0
There was 1 failed login attempt since the last successful login.


Welcome to 4.19.90-2110.8.0.0119.oe1.aarch64

System information as of time:  Wed Apr  2 15:29:36 CST 2025

System load:    0.09
Processes:      145
Memory used:    24.3%
Swap used:      0.0%
Usage On:       15%
IP address:     192.168.0.53
Users online:   1
```

4. 启动数据库服务

```
[omm@ecs-3ee7 ~]$ gs_om  -t  start
```

```
[omm@ecs-419c ~]$ gs_om  -t  start
Starting cluster.
=========================================
[SUCCESS] ecs-419c:
[2025-04-02 15:31:59.648][6781][][gs_ctl]: gs_ctl started,datadir is /gaussdb/data/db1
[2025-04-02 15:31:59.651][6781][][gs_ctl]:  another server might be running; Please use the restart command
=========================================
Successfully started.
```

5. 连接数据库

```
[omm@db1 ~]$ gsql -d postgres -p 26000 -r
```

```
openGauss=#
```

6. 创建数据库用户（学生姓名首字母：假设为 LH）

```
openGauss=# create user lxyxl with password 'lxy**lxy123';
```

```
openGauss=# create user lxyxl with password 'lxy**lxy123';
CREATE ROLE
```

7. 创建数据库并指定所有者为第 6 步新建用户 LH

```
openGauss=# CREATE DATABASE DB_Bank OWNER lxyxl;
```

```
openGauss=# CREATE DATABASE DB_Bank OWNER lxyxl;
CREATE DATABASE
```

8. 给新建用户 LH 授权使其拥有 sysadmin 权限

```
openGauss=# GRANT ALL PRIVILEGES ON DATABASE DB_Bank TO lxyxl;
openGauss=# ALTER USER lxyxl SYSADMIN;
```

```
openGauss=# GRANT ALL PRIVILEGES ON DATABASE DB_Bank TO lxyxl;
GRANT
openGauss=# ALTER USER lxyxl SYSADMIN;
ALTER ROLE
```

9. 查看当前用户（current_user）

```
openGauss=# SELECT current_user;
```

```
openGauss=# SELECT current_user;
 current_user
--------------
 omm
(1 row)
```

```
openGauss=# \du
                                                                List of roles
 Role name |                                                      Attributes
                                   | Member of
-----------+-----------------------------------------------------------------------------------
---------------------------------+-----------
 alexi     |
                                   | {}
 lxy       | Sysadmin
                                   | {}
 lxyxl     | Sysadmin
                                   | {}
 omm       | Sysadmin, Create role, Create DB, Replication, Administer audit, Monitoradmin,
Operatoradmin, Policyadmin, UseFT | {}
```

```
[omm@ecs-419c ~]$ gsql -d postgres -p 26000 -U lxyxl
Password for user lxyxl:
gsql ((openGauss 2.1.0 build 590b0f8e) compiled at 2021-09-30 14:29:27 commit 0 last mr  )
Non-SSL connection (SSL connection is recommended when requiring high-security)
Type "help" for help.

openGauss=>
```

```
openGauss=> select current_user;
 current_user
--------------
 lxyxl
(1 row)
```

10. ！查看当前模式（确保在模式 LH 下）

```
openGauss=# alter session set current_schema=lsh;
openGauss=> select current_schema();
```

```
openGauss=> alter session set current_schema=lsh;
SET
openGauss=> select current_schema();
 current_schema
----------------

(1 row)
```

11. 切换到数据库 ben_ben

```
openGauss=>   \c ben_ben;
```

```
db_bank->  \c ben_ben;
Password for user lxyxl:
Non-SSL connection (SSL connection is recommended when requiring high-security)
You are now connected to database "ben_ben" as user "lxyxl".
ben_ben->
```

12. 修改表结构，添加约束。给职级表 rank 的 rankid 添加主键约束（alter table 表名 add constraint 约束名称 primary key(列名)）

```
ben_ben=# SELECT *
ben_ben-# FROM information_schema.table_constraints
ben_ben-# WHERE table_name = 'rank';
ben_ben=# ALTER TABLE rank DROP CONSTRAINT rank_pkey;
ben_ben=# ALTER TABLE rank ADD CONSTRAINT lxy_rank PRIMARY KEY (rankid);
```

查询表中现有约束，发现已有主键约束 rank_pkey：

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'rank';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | rank_pkey             | ben_ben       | public       | rank       | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16399_1_not_null | ben_ben       | public       | rank       | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16399_2_not_null | ben_ben       | public       | rank       | CHECK           | NO            |
 NO
(3 rows)
```

将其删除重新添加：

```
ben_ben=# ALTER TABLE rank DROP CONSTRAINT rank_pkey;
ALTER TABLE
```

删除成功：

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'rank';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_sche
ma | table_name | constraint_type | is_deferrable | initially_deferred
--------------------+-------------------+-----------------------+---------------+-----------
---+------------+-----------------+---------------+--------------------
 ben_ben            | public            | 2200_16399_1_not_null | ben_ben       | public
   | rank       | CHECK           | NO            | NO
 ben_ben            | public            | 2200_16399_2_not_null | ben_ben       | public
   | rank       | CHECK           | NO            | NO
(2 rows)
```

添加主键约束 lxy_rank 成功：

```
ben_ben=# ALTER TABLE rank ADD CONSTRAINT lxy_rank PRIMARY KEY (rankid);
NOTICE:  ALTER TABLE / ADD PRIMARY KEY will create implicit index "lxy_rank" for table "rank"
ALTER TABLE
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'rank';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_sche
ma | table_name | constraint_type | is_deferrable | initially_deferred
--------------------+-------------------+-----------------------+---------------+-----------
---+------------+-----------------+---------------+--------------------
 ben_ben            | public            | lxy_rank              | ben_ben       | public
   | rank       | PRIMARY KEY     | NO            | NO
 ben_ben            | public            | 2200_16399_1_not_null | ben_ben       | public
   | rank       | CHECK           | NO            | NO
 ben_ben            | public            | 2200_16399_2_not_null | ben_ben       | public
   | rank       | CHECK           | NO            | NO
(3 rows)
```

13. 给部门表 department 的 departmentid 添加主键约束

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'department';
ben_ben=# ALTER TABLE department DROP CONSTRAINT department_pkey;
ben_ben=# ALTER TABLE department ADD CONSTRAINT lxy_department PRIMARY KEY (departmentid);
```

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'department';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | department_pkey       | ben_ben       | public       | department | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16407_1_not_null | ben_ben       | public       | department | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16407_2_not_null | ben_ben       | public       | department | CHECK           | NO            |
 NO
(3 rows)

ben_ben=# ALTER TABLE department DROP CONSTRAINT department_pkey;
ALTER TABLE
ben_ben=# ALTER TABLE department ADD CONSTRAINT lxy_department PRIMARY KEY (departmentid);
NOTICE:  ALTER TABLE / ADD PRIMARY KEY will create implicit index "lxy_department" for table "department"
ALTER TABLE
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'department';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | lxy_department        | ben_ben       | public       | department | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16407_1_not_null | ben_ben       | public       | department | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16407_2_not_null | ben_ben       | public       | department | CHECK           | NO            |
 NO
(3 rows)
```

14. 给职员表 people 的 peopleID 添加主键约束

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'people';
ben_ben=# ALTER TABLE people DROP CONSTRAINT people_pkey;
ben_ben=# ALTER TABLE people ADD CONSTRAINT lxy_people PRIMARY KEY (peopleid);
```

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'people';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | people_pkey           | ben_ben       | public       | people     | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16415_1_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_2_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_3_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_4_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_5_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_6_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_7_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_8_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
(9 rows)

ben_ben=# ALTER TABLE people DROP CONSTRAINT people_pkey;
ALTER TABLE
ben_ben=# ALTER TABLE people ADD CONSTRAINT lxy_people PRIMARY KEY (peopleid);
NOTICE:  ALTER TABLE / ADD PRIMARY KEY will create implicit index "lxy_people" for table "people"
ALTER TABLE
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'people';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | lxy_people            | ben_ben       | public       | people     | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16415_1_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_2_not_null | ben_ben       | public       | people     | CHECK           | NO            |
```

15. 给 people 表 peopleSex 列添加默认约束(录入数据时默认为“女”提示: alter table 表名 alter 列名 set default 默认值)

```
ben_ben=# ALTER TABLE people ALTER COLUMN peopleSex SET DEFAULT '女';
```

```
ben_ben=# ALTER TABLE people ALTER COLUMN peopleSex SET DEFAULT '女';
ALTER TABLE
```

16. 给 people 表的 peoplesalary 列添加 check 约束(薪资在 1000~100000 之间)

```
ben_ben=# ALTER TABLE people ADD CONSTRAINT ck_salary CHECK (peoplesalary BETWEEN 1000 AND 100000);
```

```
ben_ben=# ALTER TABLE people ADD CONSTRAINT ck_salary CHECK (peoplesalary BETWEEN 1000 AND 100000);
ALTER TABLE
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'people';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------
--------------------
 ben_ben            | public            | ck_salary             | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | lxy_people            | ben_ben       | public       | people     | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16415_1_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_2_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_3_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_4_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_5_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_6_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_7_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_8_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
(10 rows)
```

17. 给 people 表添加一列 Mphone(Mphone integer)

```
ben_ben=# ALTER TABLE people ADD COLUMN Mphone INTEGER;
```

```
ben_ben=# ALTER TABLE people ADD COLUMN Mphone INTEGER;
ALTER TABLE
```

```
ben_ben=# select mphone from people;
 mphone
--------
(24 rows)
```

18. 给 people 表的 Mphone 添加 unique 约束（要求每个职员的手机号不完全相同 alter table 表名 add constraint 约束名称 unique(列名)）

```
ben_ben=# alter table people add constraint uni_mphone unique(mphone);
```

```
ben_ben=# alter table people add constraint uni_mphone unique(mphone)
ben_ben-# ;
NOTICE:  ALTER TABLE / ADD UNIQUE will create implicit index "uni_mphone" for table "people"
ALTER TABLE
```

```
ben_ben=# SELECT *
FROM information_schema.table_constraints
WHERE table_name = 'people';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | uni_mphone            | ben_ben       | public       | people     | UNIQUE          | NO            |
 NO
```

19. !给部门表 departmentr 的列 RoomNum 办公室房间号添加非空约束。ALTER TABLE 表名 ALTER COLUMN 列名 SET NOT NULL

由于 department 里面没有 roomnum 这一列：

```
ben_ben=# select * from department
ben_ben-# ;
 departmentid | departmentname | departmentremark
--------------+----------------+------------------
            1 | 软件部         | ...
            2 | 硬件部         | ...
            3 | 市场部         | ...
            4 | 财务部         | 企业财务结算
            5 | 人力部         | 人事管理
            6 | 品控部         | 质量检控
(6 rows)
```

这里我们选择手动添加之后再进行后续操作：

```
ben_ben=# alter table department add column roomnum varchar(10);
ALTER TABLE
ben_ben=# select * from department
;
 departmentid | departmentname | departmentremark | roomnum
--------------+----------------+------------------+---------
            1 | 软件部         | ...              |
            2 | 硬件部         | ...              |
            3 | 市场部         | ...              |
            4 | 财务部         | 企业财务结算     |
            5 | 人力部         | 人事管理         |
            6 | 品控部         | 质量检控         |
(6 rows)
```

```
ben_ben=# update department set roomnum = case
when departmentid = 1 then '301'
when departmentid = 2 then '303'
when departmentid = 3 then '9'
when departmentid = 4 then '7'
when departmentid = 5 then '23'
when departmentid = 6 then '2004'
else roomnum
end;
UPDATE 6
ben_ben=# select * from department
;
 departmentid | departmentname | departmentremark | roomnum
--------------+----------------+------------------+---------
            1 | 软件部         | ...              | 301
            2 | 硬件部         | ...              | 303
            3 | 市场部         | ...              | 9
            4 | 财务部         | 企业财务结算     | 7
            5 | 人力部         | 人事管理         | 23
            6 | 品控部         | 质量检控         | 2004
(6 rows)
```

添加非空约束：

```
ben_ben=# alter table department alter column roomnum set not null;
ALTER TABLE
```

验证：添加新行，如果房间号为空创建失败。

```
ben_ben=# insert into department(departmentid , departmentname) values('7','新部门');
ERROR:  null value in column "roomnum" violates not-null constraint
DETAIL:  Failing row contains (7, 新部门, null, null).
```

```
ben_ben=# insert into department(departmentid , departmentname,roomnum) values('7','新部门',
'210');
INSERT 0 1
ben_ben=# select * from department
;
 departmentid | departmentname | departmentremark | roomnum
--------------+----------------+------------------+---------
            1 | 软件部         | ...              | 301
            2 | 硬件部         | ...              | 303
            3 | 市场部         | ...              | 9
            4 | 财务部         | 企业财务结算     | 7
            5 | 人力部         | 人事管理         | 23
            6 | 品控部         | 质量检控         | 2004
            7 | 新部门         |                  | 210
(7 rows)
```

20. ！删除部门表 department 的列 RoomNum 办公室房间号的非空约束

ALTER TABLE ma.student ALTER COLUMN contract_way DROP NOT NULL

```
ben_ben=# alter table department alter column roomnum drop not null
```

先把刚刚创建的行删除：

```
ben_ben=# delete from department
ben_ben-# where departmentid = 7;
DELETE 1
ben_ben=# select * from department;
 departmentid | departmentname | departmentremark | roomnum
--------------+----------------+------------------+---------
            1 | 软件部         | ...              | 301
            2 | 硬件部         | ...              | 303
            3 | 市场部         | ...              | 9
            4 | 财务部         | 企业财务结算     | 7
            5 | 人力部         | 人事管理         | 23
            6 | 品控部         | 质量检控         | 2004
(6 rows)
```

删除非空约束：

```
ben_ben=# alter table department alter column roomnum drop not null
ben_ben-# ;
ALTER TABLE
```

验证： 房间号为空时依然可以创建新的行。

```
ben_ben=# insert into department(departmentid , departmentname)
ben_ben-# values('7','新部门');
INSERT 0 1
ben_ben=# select * from department;
 departmentid | departmentname | departmentremark | roomnum
--------------+----------------+------------------+---------
            1 | 软件部         | ...              | 301
            2 | 硬件部         | ...              | 303
            3 | 市场部         | ...              | 9
            4 | 财务部         | 企业财务结算     | 7
            5 | 人力部         | 人事管理         | 23
            6 | 品控部         | 质量检控         | 2004
            7 | 新部门         |                  |
(7 rows)
```

21. ！修改部门表 department 的 RoomNum 列的数据类型为 integer。ALTER TABLE table_name ALTER COLUMN column_name type datatype

```
ben_ben=# alter table department alter column roomnum type integer;
```

```
ben_ben=# alter table department alter column roomnum type integer;
ALTER TABLE
```

22. ！删除部门表 department 的 RoomNum 列

```
ben_ben=# alter table department drop column roomnum;
```

```
ben_ben=# alter table department drop column roomnum;
ALTER TABLE
ben_ben=# select* from department;
 departmentid | departmentname | departmentremark
--------------+----------------+------------------
            1 | 软件部         | ...
            2 | 硬件部         | ...
            3 | 市场部         | ...
            4 | 财务部         | 企业财务结算
            5 | 人力部         | 人事管理
            6 | 品控部         | 质量检控
(6 rows)
```

23. 给 people 表的 departmentid、rankid 添加外键约束 alter table 表名 add constraint 约束名称 foreign key（列名）references 关联表名（关联表列名）

```
ben_ben=# ALTER TABLE people
ben_ben-#   ADD CONSTRAINT fk_department FOREIGN KEY (departmentid) REFERENCES department(departmentid),
ben_ben-#   ADD CONSTRAINT fk_rank FOREIGN KEY (rankid) REFERENCES rank(rankid);
```

```
ben_ben=# ALTER TABLE people
ben_ben-#   ADD CONSTRAINT fk_department FOREIGN KEY (departmentid) REFERENCES department(de
partmentid),
ben_ben-#   ADD CONSTRAINT fk_rank FOREIGN KEY (rankid) REFERENCES rank(rankid);
ALTER TABLE
```

```
ben_ben=# select *
from information_schema.table_constraints
where table_name = 'people';
 constraint_catalog | constraint_schema |    constraint_name    | table_catalog | table_schema | table_name | constraint_type | is_deferrable |
 initially_deferred
--------------------+-------------------+-----------------------+---------------+--------------+------------+-----------------+---------------+
--------------------
 ben_ben            | public            | fk_rank               | ben_ben       | public       | people     | FOREIGN KEY     | NO            |
 NO
 ben_ben            | public            | fk_department         | ben_ben       | public       | people     | FOREIGN KEY     | NO            |
 NO
 ben_ben            | public            | uni_mphone            | ben_ben       | public       | people     | UNIQUE          | NO            |
 NO
 ben_ben            | public            | ck_salary             | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | lxy_people            | ben_ben       | public       | people     | PRIMARY KEY     | NO            |
 NO
 ben_ben            | public            | 2200_16415_1_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_2_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_3_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_4_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_5_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_6_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_7_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
 ben_ben            | public            | 2200_16415_8_not_null | ben_ben       | public       | people     | CHECK           | NO            |
 NO
(13 rows)
```

24. 给部门表 department 的插入 1 条记录（departmentname:创新部）

```
ben_ben=# insert into department(departmentid , departmentname) values ('7','创新部');
```

```
ben_ben=# insert into department(departmentid , departmentname) values ('7','创新部');
INSERT 0 1
ben_ben=# select * from department;
 departmentid | departmentname | departmentremark
--------------+----------------+------------------
            1 | 软件部         | ...
            2 | 硬件部         | ...
            3 | 市场部         | ...
            4 | 财务部         | 企业财务结算
            5 | 人力部         | 人事管理
            6 | 品控部         | 质量检控
            7 | 创新部         |
(7 rows)
```

25. 给职员表中每个员工薪资增加 1000 元

```
ben_ben=# update people set peoplesalary = peoplesalary + 1000;
```

```
peoplesalary |
-------------+
    50000.00 |
    40000.00 |
    25000.00 |
    12000.00 |
    18000.00 |
     9000.00 |
    10500.00 |
     8500.00 |
    16500.00 |
    15000.00 |
     9000.00 |
    18000.00 |
    17500.00 |
     9000.00 |
    11000.00 |
    12000.00 |
    13000.00 |
    13500.00 |
     8000.00 |
     5500.00 |
```

```
ben_ben=# update people set peoplesalary = peoplesalary + 1000;
UPDATE 24
ben_ben=# select peoplesalary from people;
 peoplesalary
--------------
     51000.00
     41000.00
     26000.00
     13000.00
     19000.00
     10000.00
     11500.00
      9500.00
     17500.00
     16000.00
     10000.00
     19000.00
```

26. 给职员表中最后一条记录（学生自己的姓名）薪资增加 2000 元。

```
ben_ben=# UPDATE people
ben_ben-# SET peoplesalary = peoplesalary + 2000
ben_ben-# WHERE peopleID = (SELECT peopleID FROM people ORDER BY peopleID DESC LIMIT 1);
```

```
ben_ben=# UPDATE people
ben_ben-# SET peoplesalary = peoplesalary + 2000
ben_ben-# WHERE peopleID = (SELECT peopleID FROM people ORDER BY peopleID DESC LIMIT 1);
UPDATE 1
```

```
ben_ben=# select peoplesalary from people where peopleID = (select peopleid from people order by peopleid desc limit 1);
 peoplesalary
--------------
     18000.00
(1 row)
```

27. 删除部门表 department 的创新部记录

```
ben_ben=# delete from department where departmentname = '创新部';
```

```
ben_ben=# delete from department where departmentname = '创新部';
DELETE 1
ben_ben=# select * from department;
 departmentid | departmentname | departmentremark
--------------+----------------+------------------
            1 | 软件部         | ...
            2 | 硬件部         | ...
            3 | 市场部         | ...
            4 | 财务部         | 企业财务结算
            5 | 人力部         | 人事管理
            6 | 品控部         | 质量检控
(6 rows)
```

28. 新建数据库（DBBank）

```
ben_ben=# CREATE DATABASE DBBank;
```

```
ben_ben=# CREATE DATABASE DBBank;
CREATE DATABASE
```

```
[omm@ecs-419c ~]$ gsql -d postgres -p 26000 -l
                             List of databases
   Name     | Owner | Encoding | Collate | Ctype | Access privileges
------------+-------+----------+---------+-------+-------------------
 ben_ben    | lxy   | UTF8     | C       | C     | =Tc/lxy           +
            |       |          |         |       | lxy=CTc/lxy       +
            |       |          |         |       | lxy=APm/lxy
 db_bank    | lxyxl | UTF8     | C       | C     | =Tc/lxyxl         +
            |       |          |         |       | lxyxl=CTc/lxyxl   +
            |       |          |         |       | lxyxl=APm/lxyxl
 db_tpcc    | alexi | UTF8     | C       | C     |
 db_tpcc02  | omm   | UTF8     | C       | C     |
 dbbank     | omm   | UTF8     | C       | C     |
 postgres   | omm   | UTF8     | C       | C     |
 template0  | omm   | UTF8     | C       | C     | =c/omm            +
            |       |          |         |       | omm=CTc/omm
 template1  | omm   | UTF8     | C       | C     | =c/omm            +
            |       |          |         |       | omm=CTc/omm
(8 rows)
```

29. 切换到数据库 DBBank

```
openGauss=# \c dbbank;
```

```
openGauss=# \c dbbank;
Non-SSL connection (SSL connection is recommended when requiring high-security)
You are now connected to database "dbbank" as user "omm".
dbbank=#
```

30. 创建表对象，本数据库包括 5 张表，体会表间对应关系，完成表结构设计，记录在后续实验过程中添加。各表结构如下：

表设计：

表名	中文名	描述
AccountInfo	账户信息表	
BankCard	银行卡	
CardExchange	交易信息表	存储存钱和取钱的记录
CardTransfer	转账信息表	存储转账信息记录
CardStateChange	状态信息变化记录表	存储银行卡状态:1:正常,2:挂失,3:冻结,4:注销

表结构设计：

AccountInfo（账户信息表）

字段名	中文名	数据类型	描述
AccountId	账户编号	整数	主键，顺序编号
AccountCode	身份证号码	字符串	
AccountPhone	电话号码	字符串	
RealName	真实姓名	字符串	
OpenTime	开户时间	时间	

BankCard（银行卡）

字段名	中文名	数据类型	描述
CardNo	银行卡卡号	字符串	主键
AccountId	账户编号	整数	与账户信息表形成主外键关系
CardPwd	银行卡密码	字符串	

CardMoney	银行卡余额	货币	money
CardState	银行卡状态	整数	1:正常,2:挂失,3:冻结,4:注销
CardTime	开卡时间	时间	默认当前时间

CardExchange（交易信息表）

字段名	**中文名**	**数据类型**	**描述**
ExchangeId	交易编号	整数	主键，顺序编号
CardNo	银行卡号	字符串	
MoneyInBank	存钱金额	货币	
MoneyOutBank	取钱金额	货币	
ExchangeTime	交易时间	时间	

CardTransfer（转账信息表）

字段名	**中文名**	**数据类型**	**描述**
TransferId	交易编号	整数	主键，顺序编号
CardNoOut	转出银行卡号	字符串	
CardNoIn	转入银行卡号	字符串	
TransferMoney	交易金额	货币	
TransferTime	交易时间	时间	

CardStateChange（状态信息变化记录表）

字段名	**中文名**	**数据类型**	**描述**
StateId	编号	整数	主键，顺序编号
CardNo	银行卡号	字符串	
OldState	原始状态	整数	1:正常,2:挂失,3:冻结,4:注销
NewState	新状态	整数	1:正常,2:挂失,3:冻结,4:注销
StateWhy	状态变化原因	字符串	
StateTime	记录时间	时间	

说明：按照上述表格要求，建好表结构即可，记录在后续实验过程中添加。

```
dbbank=# create table Account_Info(
dbbank(# AccountId serial primary key,
dbbank(# AccountCode varchar(18) not null,
dbbank(# AccountPhone varchar(11),
dbbank(# RealName varchar(50) not null,
dbbank(# OpenTime timestamp
dbbank(# );

dbbank=# create table bankcard(
cardno varchar(20) primary key,
```

```
accountid integer references account_info(accountid),
cardpwd varchar(64) not null,
cardmoney money,
cardstate integer check(cardstate between 1 and 4),
cardtime timestamp default current_timestamp
);

dbbank=# create table cardexchange(
dbbank(# exchangeid serial primary key,
dbbank(# cardno varchar(20),
dbbank(# moneyinbank money,
dbbank(# moneyoutbank money,
dbbank(# exchangetime timestamp
dbbank(# );

dbbank=# create table cardtransfer(
transferid serial primary key,
cardnoout varchar(20),
cardnoin varchar(20),
transfermoney money,
transfertime timestamp
);

dbbank=# create table cardstatechange(
dbbank(# stateid serial primary key,
dbbank(# cardno varchar(20),
dbbank(# oldstate integer check(oldstate between 1 and 4),
dbbank(# newstate integer check(newstate between 1 and 4),
dbbank(# statewhy varchar(500),
dbbank(# statetime timestamp
dbbank(# );
```

表一：

```
dbbank=# create table Account_Info(
dbbank(# AccountId serial primary key,
dbbank(# AccountCode varchar(18) not null,
dbbank(# AccountPhone varchar(11),
dbbank(# RealName varchar(50) not null,
dbbank(# OpenTime timestamp
dbbank(# );
NOTICE:  CREATE TABLE will create implicit sequence "account_info_accoun
tid_seq" for serial column "account_info.accountid"
NOTICE:  CREATE TABLE / PRIMARY KEY will create implicit index "account_
info_pkey" for table "account_info"
CREATE TABLE
dbbank=# comment on table account_info is '账户信息表';
COMMENT
dbbank=# comment on column account_info.accountid is '账户编号';
COMMENT
dbbank=# comment on column account_info.accountcode is '身份证号码';
COMMENT
dbbank=# comment on column account_info.accountphone is '电话号码';
COMMENT
dbbank=# comment on column account_info.realname is '真实姓名';
COMMENT
dbbank=# comment on column account_info.opentime is '开户时间';
COMMENT
```

表二：

```
dbbank=# create table bankcard(
cardno varchar(20) primary key,
accountid integer references account_info(accountid),
cardpwd varchar(64) not null,
cardmoney money,
cardstate integer check(cardstate between 1 and 4),
cardtime timestamp default current_timestamp
);
NOTICE:  CREATE TABLE / PRIMARY KEY will create implicit index "bankcard_pkey" for table "bankcard"
CREATE TABLE
dbbank=# comment on column bankcard.cardno is '银行卡号';
COMMENT
dbbank=# comment on column bankcard.accountid is '账户编号';
COMMENT
dbbank=# comment on column bankcard.cardpwd is '银行卡密码';
COMMENT
dbbank=# comment on column bankcard.cardmoney is '银行卡余额';
COMMENT
dbbank=# comment on column bankcard.cardstate is '银行卡状态';
COMMENT
dbbank=# comment on column bankcard.cardtime is '开卡时间';
COMMENT
```

表三：

```
dbbank=# create table cardexchange(
dbbank(# exchangeid serial primary key,
dbbank(# cardno varchar(20),
dbbank(# moneyinbank money,
dbbank(# moneyoutbank money,
dbbank(# exchangetime timestamp
dbbank(# );
NOTICE:  CREATE TABLE will create implicit sequence "cardexchange_exchangeid_seq" for serial column "cardexchange.exchangeid"
NOTICE:  CREATE TABLE / PRIMARY KEY will create implicit index "cardexchange_pkey" for table "cardexchange"
CREATE TABLE
dbbank=# comment on column cardexchange.exchangeid is '交易编号';
COMMENT
dbbank=# comment on column cardexchange.cardno is '银行卡号';
COMMENT
dbbank=# comment on column cardexchange.moneyinbank is '存钱金额';
COMMENT
dbbank=# comment on column cardexchange.moneyoutbank is '取钱金额';
COMMENT
dbbank=# comment on column cardexchange.exchangetime is '交易时间';
COMMENT
```

表四：

```
dbbank=# create table cardtransfer(
transferid serial primary key,
cardnoout varchar(20),
cardnoin varchar(20),
transfermoney money,
transfertime timestamp
);
NOTICE:  CREATE TABLE will create implicit sequence "cardtransfer_transferid_seq" for serial column "cardtransfer.transferid"
NOTICE:  CREATE TABLE / PRIMARY KEY will create implicit index "cardtransfer_pkey" for table "cardtransfer"
CREATE TABLE
dbbank=# comment on column cardtransfer.transferid is '交易编号';
COMMENT
dbbank=# comment on column cardtransfer.cardnoout is '转出银行卡号';
COMMENT
dbbank=# comment on column cardtransfer.cardnoin is '转入银行卡号';
COMMENT
dbbank=# comment on column cardtransfer.transfermoney is '交易金额';
COMMENT
dbbank=# comment on column cardtransfer.transfertime is '交易时间';
COMMENT
```

表五：

```
dbbank=# create table cardstatechange(
dbbank(# stateid serial primary key,
dbbank(# cardno varchar(20),
dbbank(# oldstate integer check(oldstate between 1 and 4),
dbbank(# newstate integer check(newstate between 1 and 4),
dbbank(# statewhy varchar(500),
dbbank(# statetime timestamp
dbbank(# );
NOTICE:  CREATE TABLE will create implicit sequence "cardstatechange_stateid_seq" for serial column "cardstatechange.stateid"
NOTICE:  CREATE TABLE / PRIMARY KEY will create implicit index "cardstatechange_pkey" for table "cardstatechange"
CREATE TABLE
dbbank=# comment on column cardstatechange.stateid is '编号';
COMMENT
dbbank=# comment on column cardstatechange.cardno is '银行卡号';
COMMENT
dbbank=# comment on column cardstatechange.oldstate is '原始状态';
COMMENT
dbbank=# comment on column cardstatechange.newstate is '新状态';
COMMENT
dbbank=# comment on column cardstatechange.statewhy is '状态变化原因';
COMMENT
dbbank=# comment on column cardstatechange.statetime is '记录时间';
COMMENT
```

31. 查看各表的结构

使用\d+ 表名 以及 select * from 表名 查看每个表结构:

表一:

```
dbbank=# select * from accountinfo;
 accountid | accountcode | accountphone | realname | opentime
-----------+-------------+--------------+----------+----------
(0 rows)

dbbank=# \d+ accountinfo;
                                                          Table "public.accountinfo"
    Column     |            Type             |                          Modifiers                          | Storage  | Stats target | Description
-------------+----------------------------+-----------------------------------------------------------+----------+--------------+-------------
 accountid    | integer                     | not null default nextval('account_info_accountid_seq'::regclass) | plain    |              | 账户编号
 accountcode  | character varying(18)       | not null                                                  | extended |              | 身份证号码
 accountphone | character varying(11)       |                                                           | extended |              | 电话号码
 realname     | character varying(50)       | not null                                                  | extended |              | 真实姓名
 opentime     | timestamp without time zone |                                                           | plain    |              | 开户时间
Indexes:
    "account_info_pkey" PRIMARY KEY, btree (accountid) TABLESPACE pg_default
Referenced by:
    TABLE "bankcard" CONSTRAINT "bankcard_accountid_fkey" FOREIGN KEY (accountid) REFERENCES accountinfo(accountid)
Has OIDs: no
Options: orientation=row, compression=no
```

表二:

```
dbbank=# select * from bankcard;
 cardno | accountid | cardpwd | cardmoney | cardstate | cardtime
--------+-----------+---------+-----------+-----------+----------
(0 rows)

dbbank=# \d+ bankcard;
                                          Table "public.bankcard"
  Column    |            Type             |         Modifiers         | Storage  | Stats target | Description
-----------+----------------------------+---------------------------+----------+--------------+-------------
 cardno    | character varying(20)       | not null                  | extended |              | 银行卡号
 accountid | integer                     |                           | plain    |              | 账户编号
 cardpwd   | character varying(64)       | not null                  | extended |              | 银行卡密码
 cardmoney | money                       |                           | plain    |              | 银行卡余额
 cardstate | integer                     |                           | plain    |              | 银行卡状态
 cardtime  | timestamp without time zone | default pg_systimestamp() | plain    |              | 开卡时间
Indexes:
    "bankcard_pkey" PRIMARY KEY, btree (cardno) TABLESPACE pg_default
Check constraints:
    "bankcard_cardstate_check" CHECK (cardstate >= 1 AND cardstate <= 4)
Foreign-key constraints:
    "bankcard_accountid_fkey" FOREIGN KEY (accountid) REFERENCES account_info(accountid)
Has OIDs: no
Options: orientation=row, compression=no
```

表三:

```
dbbank=# select * from cardexchange;
 exchangeid | cardno | moneyinbank | moneyoutbank | exchangetime
------------+--------+-------------+--------------+--------------
(0 rows)
```

```
dbbank=# \d+ cardexchange;
                                                                Table "public.cardexchange"
    Column     |            Type             |                          Modifiers                          | Storage  | Stats target | Description
---------------+-----------------------------+-------------------------------------------------------------+----------+--------------+-------------
 exchangeid    | integer                     | not null default nextval('cardexchange_exchangeid_seq'::regclass) | plain    |              | 交易编号
 cardno        | character varying(20)       |                                                             | extended |              | 银行卡号
 moneyinbank   | money                       |                                                             | plain    |              | 存钱金额
 moneyoutbank  | money                       |                                                             | plain    |              | 取钱金额
 exchangetime  | timestamp without time zone |                                                             | plain    |              | 交易时间
Indexes:
    "cardexchange_pkey" PRIMARY KEY, btree (exchangeid) TABLESPACE pg_default
Has OIDs: no
Options: orientation=row, compression=no
```

表四：

```
dbbank=# select * from cardtransfer;
 transferid | cardnoout | cardnoin | transfermoney | transfertime
------------+-----------+----------+---------------+--------------
(0 rows)
dbbank=# \d+ cardtransfer;
                                                                Table "public.cardtransfer"
    Column     |            Type             |                          Modifiers                          | Storage  | Stats target | Description
---------------+-----------------------------+-------------------------------------------------------------+----------+--------------+-------------
 transferid    | integer                     | not null default nextval('cardtransfer_transferid_seq'::regclass) | plain    |              | 交易编号
 cardnoout     | character varying(20)       |                                                             | extended |              | 转出银行卡号
 cardnoin      | character varying(20)       |                                                             | extended |              | 转入银行卡号
 transfermoney | money                       |                                                             | plain    |              | 交易金额
 transfertime  | timestamp without time zone |                                                             | plain    |              | 交易时间
Indexes:
    "cardtransfer_pkey" PRIMARY KEY, btree (transferid) TABLESPACE pg_default
Has OIDs: no
Options: orientation=row, compression=no
```

表五：

```
dbbank=# select * from cardstatechange;
 stateid | cardno | oldstate | newstate | statewhy | statetime
---------+--------+----------+----------+----------+-----------
(0 rows)
dbbank=# \d+ cardstatechange;
                                                                Table "public.cardstatechange"
  Column   |            Type             |                          Modifiers                          | Storage  | Stats target | Description
-----------+-----------------------------+-------------------------------------------------------------+----------+--------------+-------------
 stateid   | integer                     | not null default nextval('cardstatechange_stateid_seq'::regclass) | plain    |              | 编号
 cardno    | character varying(20)       |                                                             | extended |              | 银行卡号
 oldstate  | integer                     |                                                             | plain    |              | 原始状态
 newstate  | integer                     |                                                             | plain    |              | 新状态
 statewhy  | character varying(500)      |                                                             | extended |              | 状态变化原因
 statetime | timestamp without time zone |                                                             | plain    |              | 记录时间
Indexes:
    "cardstatechange_pkey" PRIMARY KEY, btree (stateid) TABLESPACE pg_default
Check constraints:
    "cardstatechange_newstate_check" CHECK (newstate >= 1 AND newstate <= 4)
    "cardstatechange_oldstate_check" CHECK (oldstate >= 1 AND oldstate <= 4)
Has OIDs: no
Options: orientation=row, compression=no
```

查看所有的表：

```
dbbank=# \d+
                                                   List of relations
 Schema |            Name             |   Type   | Owner |    Size    |              Storage              |    Description
--------+-----------------------------+----------+-------+------------+-----------------------------------+--------------------
 public | account_info_accountid_seq  | sequence | omm   | 8192 bytes |                                   |
 public | accountinfo                 | table    | omm   | 0 bytes    | {orientation=row,compression=no}  | 账户信息表
 public | bankcard                    | table    | omm   | 0 bytes    | {orientation=row,compression=no}  | 银行卡
 public | cardexchange                | table    | omm   | 0 bytes    | {orientation=row,compression=no}  | 交易信息表
 public | cardexchange_exchangeid_seq | sequence | omm   | 8192 bytes |                                   |
 public | cardstatechange             | table    | omm   | 8192 bytes | {orientation=row,compression=no}  | 状态信息变化记录表
 public | cardstatechange_stateid_seq | sequence | omm   | 8192 bytes |                                   |
 public | cardtransfer                | table    | omm   | 0 bytes    | {orientation=row,compression=no}  | 转账信息表
 public | cardtransfer_transferid_seq | sequence | omm   | 8192 bytes |                                   |
(9 rows)
```

1）实验过程中遇到的问题及解决方法：

1. 添加非空约束时发现目标列不存在

在步骤 19 中，尝试为 department 表的 RoomNum 列添加非空约束时，发现该列未被定义。

解决方法：

1. 使用 ALTER TABLE department ADD COLUMN RoomNum VARCHAR(20)；手动添加该列。

2. 重新执行非空约束添加命令：ALTER TABLE department ALTER COLUMN RoomNum SET NOT NULL;

2. 主键约束添加失败

在步骤 14 中，为 people 表的 peopleid 添加主键约束时，因关联表的主键冲突导致失败。 因为此时已存在一个主键 people_pkey。

解决方法：

1. 删掉已存主键 ALTER TABLE people DROP CONSTRAINT people_pkey;
2. 重新添加主键，并检查所有约束。

2) 实验心得

通过本次实验,我深入掌握了 OpenGauss 数据库的 SQL 语言基础及核心操作，包括：

通过创建、修改表结构及添加约束（主键、外键、CHECK、UNIQUE 等），理解了数据库对象管理的核心逻辑。能够灵活运用 DDL（create 、alter、drop)、DML(insert 、 delete 、select 、 update)。

问题排查能力得到了提升。在添加约束或修改表结构时，遇到列缺失、外键冲突等问题，通过查看系统表（如 information_schema.table_constraints）和逐步验证，提升了调试与解决实际问题的能力。

本次实验不仅巩固了理论知识,更让我体会到数据库设计需兼顾功能需求与性能优化。未来在复杂业务场景中，将进一步探索索引优化、事务控制等高级特性，全面提升数据库应用能力。